

Module 04: Cognition — Partially Observable Markov Decision Processes

Course: Active Inference 101 | **Unit:** Mathematical Frameworks | **Audience:** First-semester undergraduates

Learning Objectives

1. Define the **POMDP** (Partially Observable Markov Decision Process) as the mathematical framework for decision-making under uncertainty.
2. Identify the five components of a POMDP: **A** (observation), **B** (transition), **C** (preferences), **D** (initial state), and **policies** π .
3. Explain how cognition involves jointly inferring hidden states and selecting actions.

Introduction

The brain doesn't just perceive — it also decides and acts. To formalize this, we need a framework that handles both uncertainty (partial observability) and decision-making. This is the POMDP.

Key Concepts

1. From HMMs to POMDPs

An HMM (Module 01) models passive observation. A **POMDP** adds agency — the ability to choose actions that change the world:

HMM: $P(o_{1:T}, s_{1:T})$ — states evolve on their own **POMDP:** $P(o_{1:T}, s_{1:T} | \pi)$ — states evolve depending on the agent's policy π

2. The Five Components of a POMDP

Active Inference uses five matrices/vectors (the "A, B, C, D" formulation):

Symbol	Name	What It Encodes	Size
A	Observation model	$P(o s)$ — how states generate observations	observations $\times$ states
B	Transition model	$P(s_t s_{t-1}, a)$ — how actions change states	states $\times$ states $\times$ actions
C	Preference vector	$\log P(o)$ — which observations the agent prefers	observations $\times$ 1
D	Initial state prior	$P(s_1)$ — beliefs about the initial state	states $\times$ 1
π	Policies	Sequences of actions to evaluate	time $\times$ policies

3. The A Matrix — Observation Model

The **A matrix** maps hidden states to observations:

Example (simple weather):

```

           sunny   rainy
umbrella [ 0.1    0.9 ]
no_umb   [ 0.9    0.1 ]
```

This says: if it's sunny, 90% chance of no umbrella; if it's rainy, 90% chance of umbrella. Each column sums to 1.

4. The B Matrix — Transition Model

The **B matrix** determines how states change under each action:

Example (thermostat with actions: heat, cool, nothing):

B[:, :, heat] =				B[:, :, cool] =			
	cold	warm	hot		cold	warm	hot
cold	[0.0	0.0	0.0]	cold	[0.7	0.6	0.1]
warm	[0.8	0.5	0.2]	warm	[0.3	0.3	0.3]
hot	[0.2	0.5	0.8]	hot	[0.0	0.1	0.6]

5. The C Vector — Preferences

The **C vector** encodes which observations the agent prefers (what it values):

Example: C = [preferred temperature = warm]

```
C = [cold: -2, warm: +2, hot: -1]
```

Positive values = preferred observations. Negative = avoided. This replaces the reward function in traditional decision-making.

6. Putting It Together — Belief Updating

At each time step t:

1. **Observe** o_t
2. **Update beliefs:** $q(s_t) \propto A(o_t, :) \times B(:, :, a_{t-1}) \times q(s_{t-1})$
3. **Evaluate policies:** Score each policy π based on Expected Free Energy (Module 05)
4. **Select action:** Choose the best policy and execute its next action
5. **Repeat**

This is the perception-action loop formalized mathematically.

Summary

The POMDP provides the complete mathematical framework for Active Inference agents. Five components (A, B, C, D, π) specify everything the agent needs: how observations are generated, how actions change states, what's preferred, and what's believed initially. Cognition is the joint process of updating beliefs and evaluating policies.

Further Reading

- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press. (Chapter 5)
- Smith, R. et al. (2022). A step-by-step tutorial on active inference. *Journal of Mathematical Psychology*, 107, 102632.
- Kaelbling, L. P., Littman, M. L., & Cassandra, A. R. (1998). Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101, 99-134.